BACKGROUND

Pretrained features for math-step monitoring

When do frozen embeddings help a standard classifier?

Illustrative step

$2x = 6$, *so* $x = 3$

Current step

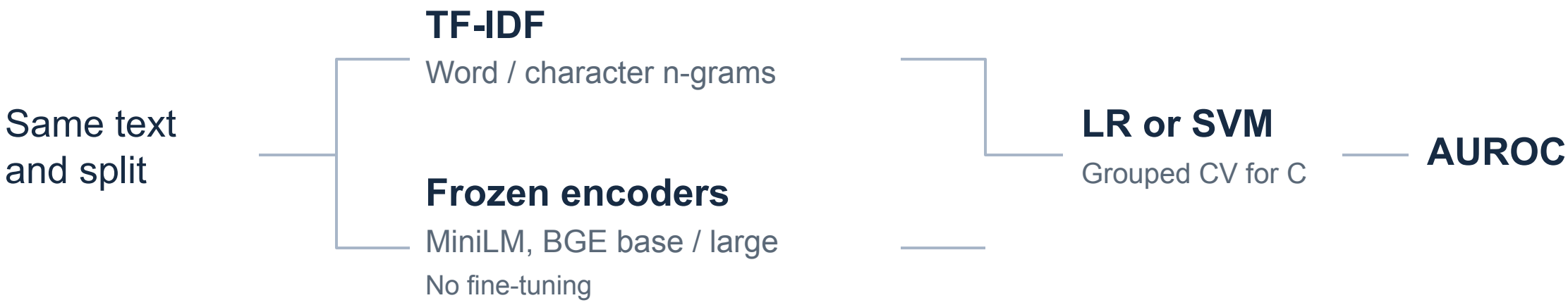
correct / incorrect

Input construction and baseline choice change the answer

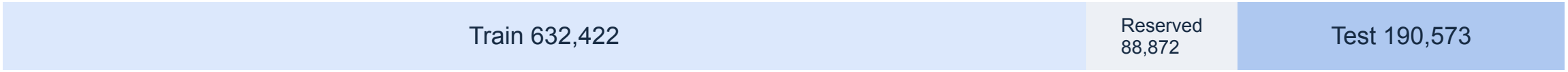
Jiayu Liao, Zeyu Pan, Yuhan Sun

PROBLEM SETUP

A controlled comparison of feature representations



PRM800K: 911,867 steps, 27.9% incorrect



Split unit = problem. Secondary metric = balanced accuracy. C = 0.1, 1, 10.

Compare features within each input condition and classifier

Sources: prepared PRM800K, src/protocol.py, src/features.py. TF-IDF inner-CV preprocessing still requires correction.

CHALLENGES

The challenge: what text should the monitor read?

Same problem split, three input constructions

Original input	Problem text + **Current step**
Position check	**Current step** + Problem text
Context removal	**Current step** only

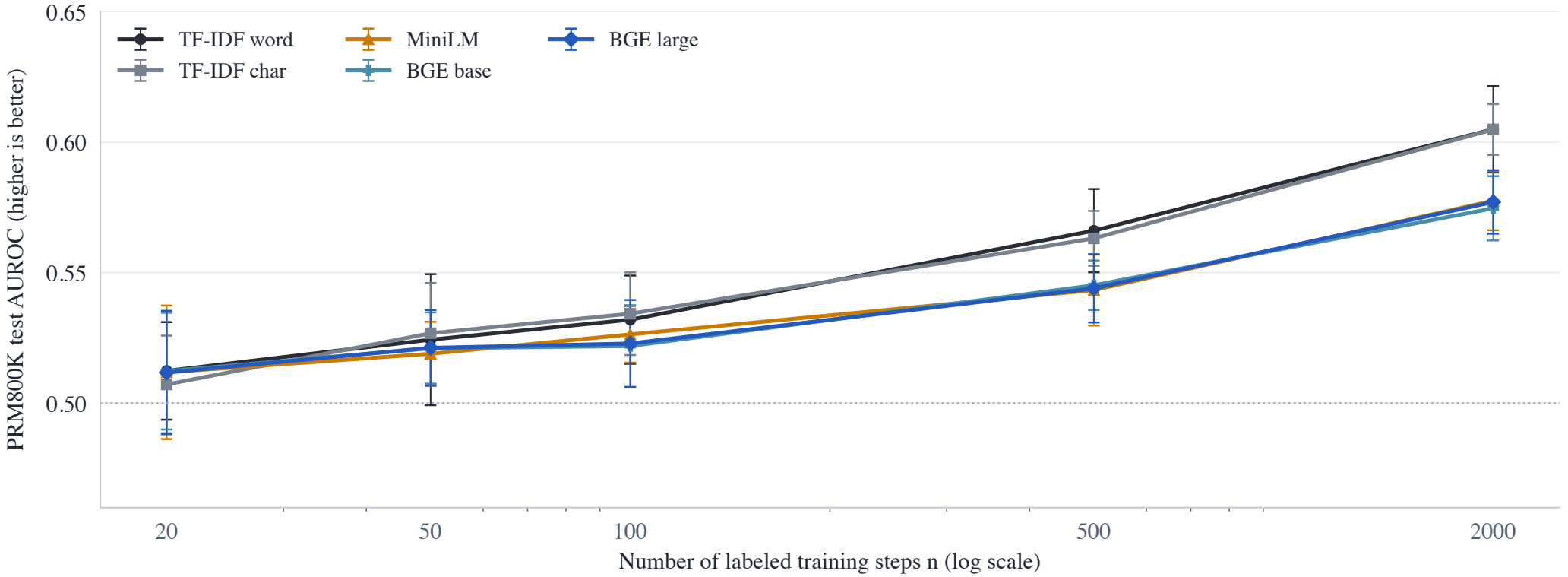
Does context help correctness prediction, or distract the representation?

Exploratory ablation. Pooling dilution is a hypothesis, not an established mechanism.

Source: input_ablation.json. The same evaluation sets have been reused during exploratory analysis.

CURRENT PROGRESS

No observed sample-efficiency gain with the original input



Problem + step input. Linear SVM. Mean ± 1 SD across 10 training subsamples, not a confidence interval.

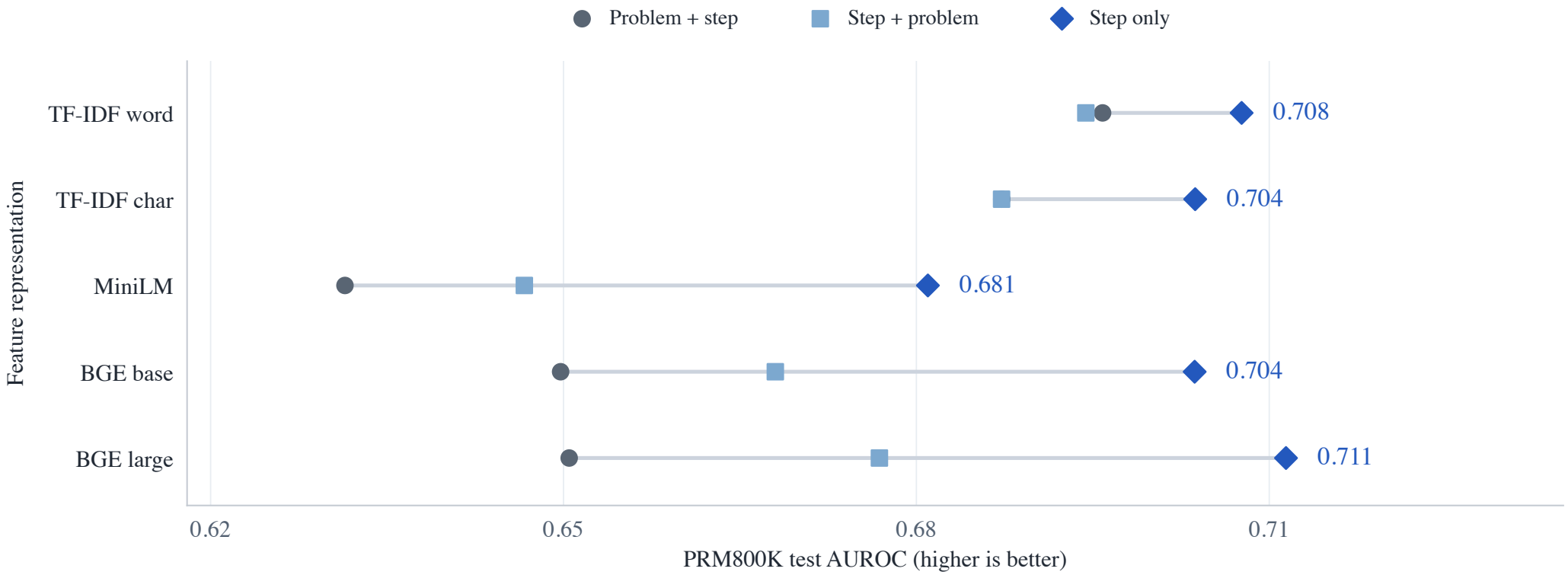
Embedding means trail word TF-IDF at n = 50 to 2,000

The step-only learning curve remains a next experiment.

Source: e1_learning_curves_prm800k.json. Subsamples vary within one fixed problem-grouped split.

CURRENT PROGRESS

Step-only input narrows the in-domain gap



Full training data, linear SVM, seed 0. One fit per condition. Point estimates without uncertainty intervals.

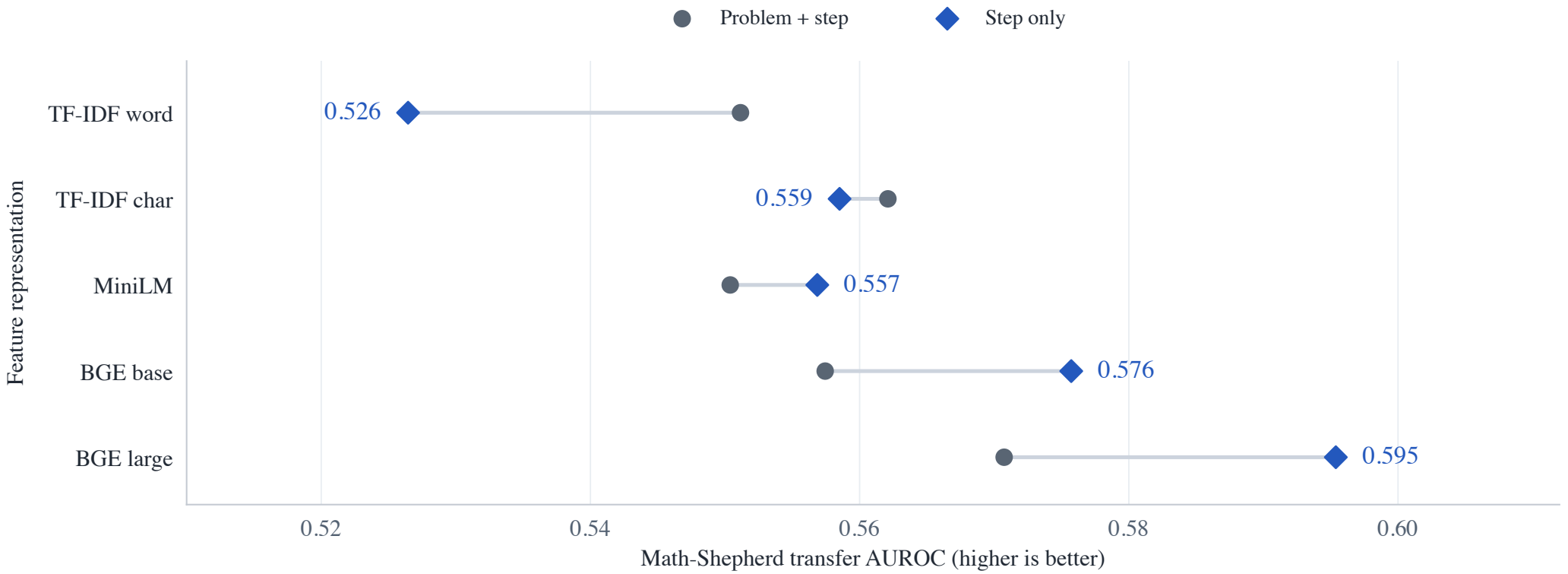
BGE large 0.711 vs word TF-IDF 0.708: numerically close

BGE truncation affects 0.04% of inputs. Context and pooling effects still need controlled tests.

Sources: input_ablation.json, truncation_check.json. No superiority or equivalence claim.

CURRENT PROGRESS

BGE transfer gains depend on the baseline



Train on PRM800K/MATH, test on Math-Shepherd/GSM8K. Full-data linear SVM, seed 0. Point estimates.

Step-only BGE large: +0.069 vs word, +0.037 vs char TF-IDF

MiniLM slightly trails char TF-IDF. The shift changes problem type, source, and labeling together.

Sources: e3_generalization.json, e3_generalization_step.json. Transfer data: 613,666 automatically labeled steps.

PLANS

Validation and the next experiments

01	**Fold-wise TF-IDF fitting**	Current vocabulary / IDF fitting precedes inner CV
02	**Problem-level paired intervals**	Current step bootstrap ignores within-problem dependence
03	**Step-only learning curves** **Separate problem / step encodings**	Test sample efficiency and the context hypothesis

Preliminary benefit is conditional on input and baseline

AI disclosure: assistants generated implementation code under group direction and helped check analysis, figures, and these materials. Group members must understand and verify the submitted work.

Current methodological limits are part of the final-stage work. Exploratory input ablation needs independent confirmation.

BACKUP 1 · Q&A ONLY

Full-data SVM results and uncertainty

Representation	PRM: problem + step	PRM: step + problem	PRM: step only	Transfer: step only
TF-IDF word	0.6958	0.6944	0.7077	0.5265
TF-IDF char	0.6873	0.6872	0.7037	0.5585
MiniLM	0.6314	0.6467	0.6810	0.5569
BGE base	0.6498	0.6680	0.7037	0.5757
BGE large	0.6505	0.6769	0.7114	0.5954

All entries = AUROC, seed 0. Input ablations use the same split and select C separately within each condition.

Existing step-only BGE-large − word TF-IDF bootstrap interval: [0.067, 0.071]. It resamples steps, not whole problems, and is conditional on one fitted model. A problem-level interval is pending. No interval against char TF-IDF is available.

Sources: input_ablation.json, e3_generalization_step.json, e3_step_transfer_delta_ci.json, src/e3_step_transfer_ci.py.

BACKUP 2 · Q&A ONLY

Small perturbation effects in the original input

AUROC drop = clean AUROC − perturbed AUROC. Positive means degradation.

Representation	Typos	Number format	LaTeX style	Combined
TF-IDF word	+0.0003	+0.0003	+0.0023	+0.0029
TF-IDF char	+0.0001	+0.0020	+0.0051	+0.0070
MiniLM	-0.0001	-0.0010	-0.0006	-0.0017
BGE base	+0.0003	+0.0001	+0.0005	+0.0009
BGE large	+0.0003	+0.0004	-0.0016	-0.0008

Linear SVM, problem + step input, seed 0. Changes are small on these rule-based tests. Different clean baselines and no step-only perturbation rerun limit interpretation.

An early label-selection bug inflated the positive-label rate. The experiments shown here use the corrected data preparation and rerun results.

Source: e4_robustness.json. Resource acknowledgments: OpenAI PRM800K, Math-Shepherd, sentence-transformers MiniLM, BAAI BGE.